```sql
-- ============================================================
--         BANKING DATABASE PROJECT
--         Presented by: [Your Group Names Here]
-- ============================================================


-- ============================================================
-- PART 1 : DATABASE + CUSTOMERS + ACCOUNTS
-- ============================================================

CREATE DATABASE BankingDB;
USE BankingDB;

-- TABLE 1: Customers
CREATE TABLE Customers (
    cust_id       INT PRIMARY KEY AUTO_INCREMENT,
    full_name     VARCHAR(100) NOT NULL,
    cust_email    VARCHAR(100) UNIQUE,
    mobile_no     VARCHAR(15),
    city          VARCHAR(100),
    birth_date    DATE
);

-- TABLE 2: Accounts
CREATE TABLE Accounts (
    acc_id        INT PRIMARY KEY AUTO_INCREMENT,
    cust_id       INT,
    acc_type      VARCHAR(20),       -- 'Savings' or 'Current'
    acc_balance   DECIMAL(10,2)  DEFAULT 0.00,
    open_date     DATE,
    FOREIGN KEY (cust_id) REFERENCES Customers(cust_id)
);
```

```sql
-- ============================================================
-- PART 2 : TRANSACTIONS + LOANS
-- ============================================================

-- TABLE 3: Transactions
CREATE TABLE Transactions (
    txn_id        INT PRIMARY KEY AUTO_INCREMENT,
    acc_id        INT,
    txn_type      VARCHAR(20),       -- 'Deposit', 'Withdrawal', 'Transfer'
    txn_amount    DECIMAL(10,2),
    txn_date      DATETIME DEFAULT NOW(),
    FOREIGN KEY (acc_id) REFERENCES Accounts(acc_id)
);

-- TABLE 4: Loans
CREATE TABLE Loans (
    loan_id       INT PRIMARY KEY AUTO_INCREMENT,
    cust_id       INT,
    loan_amount   DECIMAL(10,2),
    interest_rate DECIMAL(5,2),
    loan_date     DATE,
    loan_status   VARCHAR(20) DEFAULT 'Active',   -- 'Active' or 'Closed'
    FOREIGN KEY (cust_id) REFERENCES Customers(cust_id)
);


-- INSERT: Customers
INSERT INTO Customers (full_name, cust_email, mobile_no, city, birth_date) VALUES
('Ravi Kumar',   'ravi@gmail.com',   '9876543210', 'Hyderabad', '1990-05-15'),
('Priya Sharma', 'priya@gmail.com',  '9123456789', 'Bangalore', '1995-08-22'),
('Arjun Reddy',  'arjun@gmail.com',  '9988776655', 'Chennai',   '1988-03-10'),
('Sneha Patel',  'sneha@gmail.com',  '9871234560', 'Mumbai',    '2000-11-01'),
('Vikram Singh', 'vikram@gmail.com', '9001122334', 'Delhi',     '1985-07-19');
```

```sql
-- INSERT: Accounts
INSERT INTO Accounts (cust_id, acc_type, acc_balance, open_date) VALUES
(1, 'Savings', 15000.00, '2020-01-10'),
(2, 'Current', 50000.00, '2019-06-15'),
(3, 'Savings',  8000.00, '2021-03-20'),
(4, 'Savings', 25000.00, '2022-07-11'),
(5, 'Current', 75000.00, '2018-09-05');

-- INSERT: Transactions
INSERT INTO Transactions (acc_id, txn_type, txn_amount, txn_date) VALUES
(1, 'Deposit',     5000.00, '2024-01-05'),
(1, 'Withdrawal',  2000.00, '2024-01-10'),
(2, 'Deposit',    10000.00, '2024-01-12'),
(3, 'Withdrawal',  1500.00, '2024-02-01'),
(4, 'Deposit',     3000.00, '2024-02-14'),
(5, 'Transfer',    7000.00, '2024-03-01');

-- INSERT: Loans
INSERT INTO Loans (cust_id, loan_amount, interest_rate, loan_date, loan_status) VALUES
(1, 100000.00,  8.5, '2022-06-01', 'Active'),
(3,  50000.00, 10.0, '2023-01-15', 'Active'),
(5, 200000.00,  7.5, '2021-11-20', 'Closed');


-- ============================================================
-- PART 3 : QUERIES
-- ============================================================

-- Q1: Show all customers
SELECT * FROM Customers;

-- Q2: Show all accounts with customer names (JOIN)
SELECT c.full_name, a.acc_id, a.acc_type, a.acc_balance
FROM Customers c
JOIN Accounts a ON c.cust_id = a.cust_id;

-- Q3: Customers with balance more than 20,000
SELECT c.full_name, a.acc_balance
FROM Customers c
JOIN Accounts a ON c.cust_id = a.cust_id
WHERE a.acc_balance > 20000;
```

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-- Q4: Total balance in the bank
SELECT SUM(acc_balance) AS total_bank_balance FROM Accounts;

-- Q5: All transactions of a specific account
SELECT * FROM Transactions WHERE acc_id = 1;

-- Q6: Count of each account type
SELECT acc_type, COUNT(*) AS total_accounts
FROM Accounts
GROUP BY acc_type;
```

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-- Q7: Customers who have taken loans
SELECT c.full_name, l.loan_amount, l.interest_rate, l.loan_status
FROM Customers c
JOIN Loans l ON c.cust_id = l.cust_id;

-- Q8: Highest balance account holder
SELECT c.full_name, a.acc_balance
FROM Customers c
JOIN Accounts a ON c.cust_id = a.cust_id
ORDER BY a.acc_balance DESC
LIMIT 1;

-- Q9: Total deposits and withdrawals
SELECT txn_type, SUM(txn_amount) AS total_amount
FROM Transactions
GROUP BY txn_type;

-- Q10: Active loans only
SELECT c.full_name, l.loan_amount
FROM Customers c
JOIN Loans l ON c.cust_id = l.cust_id
WHERE l.loan_status = 'Active';

-- ============================================================
-- END OF PROJECT
-- ============================================================
```

OUTPUTS :

cust_id	full_name	cust_email	mobile_no	city	birth_date
1	Ravi Kumar	ravi@gmail.com	9876543210	Hyderabad	1990-05-15
2	Priya Sharma	priya@gmail.com	9123456789	Bangalore	1995-08-22
3	Arjun Reddy	arjun@gmail.com	9988776655	Chennai	1988-03-10
4	Sneha Patel	sneha@gmail.com	9871234560	Mumbai	2000-11-01
5	Vikram Singh	vikram@gmail.com	9001122334	Delhi	1985-07-19
NULL	NULL	NULL	NULL	NULL	NULL

full_name	acc_id	acc_type	acc_balance
Ravi Kumar	1	Savings	15000.00
Priya Sharma	2	Current	50000.00
Arjun Reddy	3	Savings	8000.00
Sneha Patel	4	Savings	25000.00
Vikram Singh	5	Current	75000.00

full_name	acc_balance
Priya Sharma	50000.00
Sneha Patel	25000.00
Vikram Singh	75000.00

total_bank_balance
173000.00

txn_id	acc_id	txn_type	txn_amount	txn_date
1	1	Deposit	5000.00	2024-01-05 00:00:00
2	1	Withdrawal	2000.00	2024-01-10 00:00:00
NULL	NULL	NULL	NULL	NULL

acc_type	total_accounts
Current	2
Savings	3

full_name	loan_amount	interest_rate	loan_status
Ravi Kumar	100000.00	8.50	Active
Arjun Reddy	50000.00	10.00	Active
Vikram Singh	200000.00	7.50	Closed

full_name	acc_balance
Vikram Singh	75000.00

txn_type	total_amount
Deposit	18000.00
Transfer	7000.00
Withdrawal	3500.00

full_name	loan_amount
Ravi Kumar	100000.00
Arjun Reddy	50000.00